

20260126EB_MN (NCT, 80, IF)

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2026年1月26日 星期一

ALL media is prepared after I start the differentiation. (I added supplement fresh. The base media is prepared at the end the last year)

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate

1. 3 of 6-wells with 10mL T2 in cell-repellent plates with ReleSR (Plate I)
2. 3 of 6-wells with 10mL T2 in cell-repellent plates with accutase (Plate II)

keep some iPSCs

2026年1月27日 星期二

Plate I has less cells with more big clusters

Plate II has more cells with smaller clusters but more single cells.

2026年1月28日 星期三

T2 -> T3

2026年1月30日 星期五

T3 -> T4 (~14 mL for over-weekend)

2026年2月2日 星期一

T4 -> T5

2026年2月4日 星期三

T5 -> T6

2026年2月6日 星期五

T6 -> T6 (~14mL for over-weekend)

2026年2月9日 星期一

T6 -> T6

2026年2月10日 星期二

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

	A	B
1	cell	passage
2	WT11	

Table1 ^

	A	B
1	run1	volume (mL)
2	trypsin	3
3	HI-FBS/CTWM	6
4	run2	*take some undissolved EB
5	trypsin	2
6	HI-FBS/CTWM	4

Table3 ✓

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	4.26	99	7	29.82	A	96	60	36	507
3					B	24	250	16	939
4					C,D	6	1972	12	the rest

PlateA ^

	1	2	3	4	5	6
A	110			80		
B	111					
C	110					
D	111					

PlateB ^

	1	2	3
A			
B			

2026年2月12日 星期四

T7 -> T7 (add virus today, see the plate layout above)

110: 2uL/well, 12uL

111: 3uL/well, 18uL

each is 6 wells in 2.5mL media (400uL each)

80: 1uL/well, 2uL in 800uL

The rest:

Plate A -> add 500uL/well

Plate B -> remove 700uL and add 1000uL per well.

2026年2月13日 星期五

T7 -> T8

2026年2月14日 星期六

1/2 T8 -> 1/2 T8

2026年2月16日 星期一

1/2 T8 -> 1/2 T8

2026年2月18日 星期三

Treat SirDNA 1:6000 for A4 and B4 at 8pm

2026年2月19日 星期四

Imaging: vLH80

Table2					
	A	B	C	D	E
1	condition#				
2	vLH80		treat	image	note
3	4	2.5mM gluc 1h	10:30	11:31	I mostly imaged the upper part of the well.
4	5	0.15mM gluc 1h	10:45	11:50	
5				12:00	repeat the control again since I felt the cells didn't look very comparable to 0.15mM group. (I imaged the lower part of the well this time)

80_O.C. , 100X, 1xzoom, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	400 ms	the signal was dim. It might becasue I decreased the virus I added.
3	640	16	10	100 ms	

fileID_treatment_80				
	A	B	C	D
1	image	treatment	treat time	
2	vLH80_MN_100x_1xzoom_0001	2.5mM gluc	1h	
3	vLH80_MN_100x_1xzoom_0002	0.15mM gluc	1h	
4	vLH80_MN_100x_1xzoom_0003	2.5mM gluc	1h	repeat the same condition (different cells) after 30min.

Treat SirDNA (1:6000) for C1-3 & D1-3 500uL*6=3000uL --> 0.5uL SirDNA +9.7uL BP T8 media --> each well 1.6uL

Treat SPY595 (1:5000) for D4-6 500*4 = 1500uL --> 0.3uL in 6uL media --> 2uL per well

2026年2月20日 星期五

Imaging NCT with recurrent hypoglycemia

- ☒ Prepare more media
- 2.5mM gluc: 50uL 1.1M in 22mL media
 - 0.15mM gluc: 3.82uL 1.1M in 28mL media

NCT_O.C. , 60X, 600x900, 16-bit						
	A	B	C	D	E	F
1	channel	bit	power (%)	exposure time	note	note2
2	561	16	20	300 ms	for 110 (I was about to set it as 30%, but I didn't successfully save it)	I switched to use 30% for 110 for 24h timepoint.
3	561	16	30	300 ms	for 111	
4	640	16	10	100 ms		
5	488	16	50	1s		

Imaging Note:

- I infected the cells two days after plating. The intensity of 110 is okay (it's not bright, but it's comparable to what I had before~130-200). The mCherry+ cells are plenty, but still need time to find cells with higher intensity for imaging. Need to use more virus or higher laser power.
- C3 is a bit hard to find cells with good expression. The intensity of most of the cells are too low.
- 111 signal is also okay. With 30% laser, there are many cells are okay to image. Easier to find a good cells than 110.

condition#		
	A	B
1	condition#	
2	1	2.5 mM > 2.5 mM 1h > 2.5 mM 3h > 2.5 mM 1h > 2.5 mM 3h > 2.5 mM 1h -> 2.5mM
3	2	2.5 mM > 0.15 (1h) > 2.5 mM 3h > 0.15 (1h) > 2.5 mM 3h > 0.15 (1h) -> 0.15mM
4	3	2.5 mM > 0.15 mM 1h > 0.15mM 3h > 0.15mM 1h > 0.15mM 3h > 0.15mM 1h -> 0.15mM
5	4	2.5mM
6	5	0.15mM
7	6	0.15mM+Ver150nM

Conditions for plateA							^
	1	2	3	4	5	6	
A	4	5	6	4	4	4	
B	4	5	6	5	5	5	
C	1	2	3				
D	1	2	3	1 (TDP43)	2 (TDP43)	3 (TDP43)	

time table for treatment																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1				treat#1			treat#2		treat#3		treat#4		treat#5				
2				0h		1h			4h		5h		8h		9h	24h	
3	condition#		condition	media	time	imaging	media after imaging	time	media	time	media	time	media	time	imaging	imaging on 2/21	
4			110														
5	1	C1	2.5mM glucose constant	2.5mM gluc	11:00	12:00	2.5mM gluc	12:20	2.5mM gluc	15:03	2.5mM gluc	16:00	2.5mM gluc	19:00	20:00	11:22	
6	2	C2	0.15mM glucose recurrent	0.15mM gluc	11:30	12:35	2.5mM gluc	12:50	0.15m M gluc	15:30	2.5mM gluc	16:30	0.15mM gluc	19:30	20:35	11:57	
7	3	C3	0.15mM glucose constant	0.15mM gluc	12:00	13:11	0.15mM gluc	13:30	0.15m M gluc	16:00	0.15mM gluc	17:00	0.15mM gluc	20:00	21:15	12:27	
8			111														
9	1	D1	2.5mM glucose constant	2.5mM gluc	12:30	13:47	2.5mM gluc	14:05	2.5mM gluc	16:35	2.5mM gluc	17:35	2.5mM gluc	20:30	22:00	13:01	
10	2	D2	0.15mM glucose recurrent	0.15mM gluc	13:00	14:15	2.5mM gluc	14:30	0.15m M gluc	17:00	2.5mM gluc	18:00	0.15mM gluc	21:08	22:27	13:37	
11	3	D3	0.15mM glucose constant	0.15mM gluc	13:30	14:41	0.15mM gluc	14:55	0.15m M gluc	17:30	0.15mM gluc	18:30	0.15mM gluc	21:30	22:53	13:30	
12	on 2/21																
13	TDP43 on 2/21/2026																
14		D4	2.5mM glucose constant	2.5mM gluc + JFX650	9:30	40:30	2.5mM gluc	40:40	2.5mM gluc	43:30	2.5mM gluc	44:30	2.5mM gluc + JFX650	47:30	40:30	40:30	(add JFX650 at 9:30)
15		D5	0.15mM glucose recurrent	0.15mM gluc + JFX650	9:35	40:35	2.5mM gluc	40:40	0.15m M gluc	43:30	2.5mM gluc	44:30	0.15mM gluc + JFX650	47:30	40:30	40:30	(add JFX650 at 9:30)
16		D6	0.15mM glucose constant	0.15mM gluc + JFX650	9:40	40:40	0.15mM gluc	40:40	0.15m M gluc	43:30	0.15mM gluc	44:30	0.15mM gluc + JFX650	47:30	40:30	40:30	(add JFX650 at 9:30)
17	6-well plate				9:50	10:50				13:45 (after imaging D3 24h)		14:45		18:10	19:05-10 extract		
18																	
19					treat	1h image	24h image on 2/22										
20	110																
21	4	A1	2.5mM	2.5mM gluc	15:00	16:00	15:11										
22	5	A2	0.15mM	0.15mM gluc	15:30	16:30	15:48										
23	6	A3	0.15mM+Ver150n M		16:00	17:00	16:14										
24	111																
25	4	B1	2.5mM		16:30	17:30	16:43										
26	5	B2	0.15mM		17:00	18:00	17:27										
27	6	B3	0.15mM+Ver150n M		17:30	18:30	18:00										
28	2/22/26																
29	IF				18:20	19:25 fix											
30																	

20260220_timetable													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1		C1		C2		C3		D1		D2		D3	
2	11:00	treat#1	2.5mM gluc										
3	11:30			treat#1	0.15mM gluc								
4	12:00	image@1h				treat#1	0.15mM gluc						
5	12:30	treat#2	2.5mM gluc	image@1h				treat#1	2.5mM gluc				
6	13:00			treat#2	2.5mM gluc	image@1h				treat#1	0.15mM gluc		
7	13:30					treat#2	0.15mM gluc	image@1h				treat#1	0.15mM gluc
8	14:00							treat#2	2.5mM gluc	image@1h			
9	14:30									treat#2	2.5mM gluc	image@1h	
10	15:00	treat#3	2.5mM gluc									treat#2	0.15mM gluc
11	15:30			treat#3	0.15mM gluc								
12	16:00	treat#4	2.5mM gluc			treat#3	0.15mM gluc						
13	16:30			treat#4	2.5mM gluc			treat#3	2.5mM gluc				
14	17:00					treat#4	0.15mM gluc			treat#3	0.15mM gluc		
15	17:30							treat#4	2.5mM gluc			treat#3	0.15mM gluc
16	18:00									treat#4	2.5mM gluc		
17	18:30											treat#4	0.15mM gluc
18	19:00	treat#5	2.5mM gluc										
19	19:30			treat#5	0.15mM gluc								
20	20:00	image@9h				treat#5	0.15mM gluc						
21	20:30			image@9h				treat#5	2.5mM gluc				
22	21:00					image@9h				treat#5	0.15mM gluc		
23	21:30							image@9h				treat#5	0.15mM gluc
24	22:00									image@9h			
25	22:30											image@9h	
26	23:00												

fileID_treatment_NCT

	A	B	C	D	E	F
1	image	reporter	treatment	time	type	
2	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	2.5mM gluc	1h	continuous	
3	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	2.5mM gluc	1h	continuous	
4	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.15mM gluc	1h	recurrent	
5	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0.15mM gluc	1h	recurrent	
6	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	0.15mM gluc	1h	continuous	
7	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	0.15mM gluc	1h	continuous	
8	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	2.5mM gluc	9h	continuous	
9	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	2.5mM gluc	9h	continuous	
10	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	0.15mM gluc	9h	recurrent	
11	com_110_MN_no-glu_50%_1s_30s_60x_0010	110	0.15mM gluc	9h	recurrent	
12	com_110_MN_no-glu_50%_1s_30s_60x_0011	110	0.15mM gluc	9h	continuous	
13	com_110_MN_no-glu_50%_1s_30s_60x_0012	110	0.15mM gluc	9h	continuous	
14	com_110_MN_no-glu_50%_1s_30s_60x_0013	110	2.5mM gluc	24h	continuous	
15	com_110_MN_no-glu_50%_1s_30s_60x_0014	110	2.5mM gluc	24h	continuous	
16	com_110_MN_no-glu_50%_1s_30s_60x_0015	110	0.15mM gluc	24h	recurrent	
17	com_110_MN_no-glu_50%_1s_30s_60x_0016	110	0.15mM gluc	24h	recurrent	
18	com_110_MN_no-glu_50%_1s_30s_60x_0017	110	0.15mM gluc	24h	continuous	
19	com_110_MN_no-glu_50%_1s_30s_60x_0018	110	0.15mM gluc	24h	continuous	
20	com_110_MN_no-glu_50%_1s_30s_60x_0019	110	2.5mM gluc	1h	no-change	
21	com_110_MN_no-glu_50%_1s_30s_60x_0020	110	2.5mM gluc	1h	no-change	
22	com_110_MN_no-glu_50%_1s_30s_60x_0021	110	0.15mM gluc	1h	no-change	
23	com_110_MN_no-glu_50%_1s_30s_60x_0022	110	0.15mM gluc	1h	no-change	
24	com_110_MN_no-glu_50%_1s_30s_60x_0023	110	0.15mM+Ver1 50nM	1h	no-change	
25	com_110_MN_no-glu_50%_1s_30s_60x_0024	110	0.15mM+Ver1 50nM	1h	no-change	
26	com_110_MN_no-glu_50%_1s_30s_60x_0025	110	2.5mM gluc	24h	no-change	
27	com_110_MN_no-glu_50%_1s_30s_60x_0026	110	2.5mM gluc	24h	no-change	
28	com_110_MN_no-glu_50%_1s_30s_60x_0027	110	0.15mM gluc	24h	no-change	
29	com_110_MN_no-glu_50%_1s_30s_60x_0028	110	0.15mM gluc	24h	no-change	
30	com_110_MN_no-glu_50%_1s_30s_60x_0029	110	0.15mM+Ver1 50nM	24h	no-change	
31	com_110_MN_no-glu_50%_1s_30s_60x_0030	110	0.15mM+Ver1 50nM	24h	no-change	
32	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	2.5mM gluc	1h	continuous	
33	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	2.5mM gluc	1h	continuous	
34	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.15mM gluc	1h	recurrent	
35	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0.15mM gluc	1h	recurrent	
36	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	0.15mM gluc	1h	continuous	

37	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	0.15mM gluc	1h	continuous	
38	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	2.5mM gluc	9h	continuous	
39	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	2.5mM gluc	9h	continuous	
40	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	0.15mM gluc	9h	recurrent	
41	com_111_MN_no-glu_50%_1s_30s_60x_0010	111	0.15mM gluc	9h	recurrent	
42	com_111_MN_no-glu_50%_1s_30s_60x_0011	111	0.15mM gluc	9h	continuous	
43	com_111_MN_no-glu_50%_1s_30s_60x_0012	111	0.15mM gluc	9h	continuous	
44	com_111_MN_no-glu_50%_1s_30s_60x_0013	111	2.5mM gluc	24h	continuous	
45	com_111_MN_no-glu_50%_1s_30s_60x_0014	111	2.5mM gluc	24h	continuous	
46	com_111_MN_no-glu_50%_1s_30s_60x_0015	111	0.15mM gluc	24h	recurrent	
47	com_111_MN_no-glu_50%_1s_30s_60x_0016	111	0.15mM gluc	24h	recurrent	
48	com_111_MN_no-glu_50%_1s_30s_60x_0017	111	0.15mM gluc	24h	continuous	
49	com_111_MN_no-glu_50%_1s_30s_60x_0018	111	0.15mM gluc	24h	continuous	
50	com_111_MN_no-glu_50%_1s_30s_60x_0019	111	2.5mM gluc	1h	no-change	
51	com_111_MN_no-glu_50%_1s_30s_60x_0020	111	2.5mM gluc	1h	no-change	
52	com_111_MN_no-glu_50%_1s_30s_60x_0021	111	0.15mM gluc	1h	no-change	
53	com_111_MN_no-glu_50%_1s_30s_60x_0022	111	0.15mM gluc	1h	no-change	
54	com_111_MN_no-glu_50%_1s_30s_60x_0023	111	0.15mM+Ver1 50nM	1h	no-change	
55	com_111_MN_no-glu_50%_1s_30s_60x_0024	111	0.15mM+Ver1 50nM	1h	no-change	
56	com_111_MN_no-glu_50%_1s_30s_60x_0025	111	2.5mM gluc	24h	no-change	
57	com_111_MN_no-glu_50%_1s_30s_60x_0026	111	2.5mM gluc	24h	no-change	
58	com_111_MN_no-glu_50%_1s_30s_60x_0027	111	0.15mM gluc	24h	no-change	
59	com_111_MN_no-glu_50%_1s_30s_60x_0028	111	0.15mM gluc	24h	no-change	
60	com_111_MN_no-glu_50%_1s_30s_60x_0029	111	0.15mM+Ver1 50nM	24h	no-change	
61	com_111_MN_no-glu_50%_1s_30s_60x_0030	111	0.15mM+Ver1 50nM	24h	no-change	

2026年2月21日 星期六

Imaging ~~TDP43~~ and RNA extraction (6-well) with recurrent hypoglycemia

Treat 1:6000 SirDNA to A1-3 and B1-3 at around 9:40am

JFX650

stock:1mM

working:1nM

1. dilute 1mM JFK650 1:400 (0.5uL in 200uL) in no-glu media
2. add 1:2500 into 2.5mM or 0.15mM glucose media
 - a. 0.48 in 1.2mL 2.5mM gluc
 - b. 0.96 in 2.4mL 0.15mM gluc

Verdineor 150nM

stock: 10mM

working: 150nM
dilute to 150uM first. (1:66.7) 1uL into 65.7uL media. Then 1:1000. 1.4uL in 1.4mL 0.15mM media.

RNA extraction collect at 9h at 7:10pm

1. remove media as much as possible

2. add Trizol or Zymo DNA/RNA Shield

a. row B use 1mL Trizol. I added Trizol to the sample first and then try to add Zymo DNA/RNA Shield. I found it was a bit hard to collect that sample so I collected Trizol sample into tube first and put it on bench when I am keep collecting samples from Row A.

b. row A use 200uL Zymo DNA/RNA Shield (trial exp)

I. note: it's super easy to have bubbles with Zymo DNA/RNA Shield. I added 50uL in A1 at the beginning and almost all of them became bubbles. Then I added 100uL for A2 and A3 and found it was still hard to rinse the well with solution. At the end, I added 100uL more solution. I used cell scrapper at the end for A2 (with bigger scraper) to collect solution, but it doesn't work well with A3 (with smaller scraper at the other end), which already had a lot of bubbles). I didn't use scrapper for A1. Only A2 I sucessfully avoid most bubbles, but A1 and A3 had a lot of bubbles. After I collect samples into tubes, I centrifuged them and found I only have around 50uL samples from A1 and A3 and have around 150uL for A2.

II. Next time, I should add 150-200uL at the beggining and avoid bubbles. Perhaps I can directly scrap cells.

3. Collect samples in the tube and freeze them in -80 at 7:40pm (I hope this won't be too long for Trizol).
- Well3 ^

	1	2	3
A	1	2	3
B	1	2	3
- Note:
TDP43
- TDP43 signal looked very weird. Not totally sure if it's because the long treatment for SPY595. I don't really see the usual phenotype and it looked like a bleedthrough from SPY595. Do not continue doing the experiment since I don't think the signal is real. Perhaps change DNA dye or add the dye on the same day. I checked several images and I didn't see the same pattern I saw today (many puncta in cytosol, which usually has more diffused phenotype)
- NCT

• I switched to 30% 561 for 110 at 24h timepoint.

• I accidentally use 111 job for C1 (110). So I will add 3 time frame right after it finished acquisition. --> I should actually add 5 frames....

• The signal of A1-3 and B1-3 is better. It might because I used 30% laser power or the cells have more time to produce reporter.

• The export rate of B3 (0.15mM with 150nM KPT-335 might not get inhibited in 1h. Perhaps I should use LMB instead.
- On this page: Entry

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20260221_timetable																				
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1		C1	C2	C3	D1	D2	D3	D4		D5		D6		6-well	A1	A2	A3	B1	B2	B3
2	9:30							treat#1	0-15mM-JFK	treat#1	0-15mM-JFK	treat#1	0-15mM-JFK	treat#1						
3	10:00							image@1h		image@1h		image@1h		treat#2						
4	10:30							treat#2	0-15mM-glucose	treat#2	0-15mM-glucose	treat#2	0-15mM-glucose	treat#2						
5	11:00	image@24h						treat#3	0-15mM-glucose	treat#3	0-15mM-glucose	treat#3	0-15mM-glucose	treat#3						
6	11:30		image@24h																	
7	12:00			image@24h																
8	12:30				image@24h															
9	13:00					image@24h														
10	13:30						image@24h	treat#3	0-15mM-glucose	treat#3	0-15mM-glucose	treat#3	0-15mM-glucose	treat#3						
11	14:00																			
12	14:30							treat#4	0-15mM-glucose	treat#4	0-15mM-glucose	treat#4	0-15mM-glucose	treat#4	treat 2.5mM					
13	15:00															treat 0.15mM				
14	15:30															image@1h	treat 2.5+MPA			
15	16:00																image@1h	treat 2.5mM		
16	16:30																	image@1h	treat 0.15mM	
17	17:00																			
18	17:30							treat#5	0-15mM-JFK	treat#5	0-15mM-JFK	treat#5	0-15mM-JFK	treat#5				image@1h		
19	18:00																		image@1h	
20	18:30							image@1h		image@1h		image@1h		collect (9h)						image@1h

2026年2月22日 星期日

Note:
NCT

- A3 (110 with NCT inhibitor): Most signal are in the nucleus, but signal still increased with activation.

20260221_timetable1								
	A	B	C	D	E	F	G	H
1		A1	A2	A3	B1	B2	B3	IF
2	15:15	image@24h						
3	15:45		image@24h					
4	16:15			image@24h				
5	16:45				image@24h			
6	17:15					image@24h		
7	17:45						image@24h	
8	18:15							treat cells
9	18:45							prepare reagents
10	19:15							fix
11								

IF

Use new 1mM JFK650

- Treat cells (A5,6 & B5,6) with different glucose + Halo-dye (10nM) for 1h at 18:20-25
 - 600uL*2 = 1200, 0.5uL 25uM in 1250uL media

2. Prepare 4%PFA (7 wells: 300uL/well. 2210uL PBS + 70uL 32%PFA) wrong calculation. (But at the end I did the calculation on site and I fixed 16wells)
3. ~~Wash cells with cold PBS at~~ Only remove the media
4. Fix cells with 4% PFA for 15 min at 19:25 (fix 16 wells marked in purple below)
5. Wash with PBS
6. Permeabilization with 0.2% triton (100uL 5% triton in 2400uL PBS) for 10min (I didn't permeabilize D1-3, but permeablize all the rest)
7. Blocking with 10% FBS for overnight in cold room.

2026年2月23日 星期一

1. Primary antibodies overnight.

IF antibody layout

^

	1	2	3	4	5	6
A					B	A
B					B	A
C					D	C
D						

Table7

/

	A	B	C	D	E	F	G	H	I	J	K	L
1	condition	Ab#	antibody	species	dilution	total volume	Ab volume	secondary Ab	#	dilution	total volume	Ab volume
2	A	33	TDP-43	Rb	500	360	0.72	488	B	1000	900	0.9
3	A	147	Importin-b1	m	1000	360	0.36	594	O	1000	900	0.9
4	B	C-31	RANGAP1	Rb	500	360	0.72	488	B			
5	B	146	Ran	m	500	360	0.72	594	O			
6	C	C-31	RANGAP1	Rb	500	360	0.72					
7	C	147	Importin-b1	m	1000	360	0.36					

antibodies for MN identity											
	A	B	C	D	E	F	G	H	I	J	K
1							dilution	total volume		secondary Ab	#
2	C	C-32	Anti-MNX1 (HB9) Antibody	ABN174	Poly, Rb	spinal motor neuron fate markers	500	180	0.36	488	B
3	C	C-33	Phox2b (B-11)	sc-376997	m, Mono	cranial motor neuron markers	500	180	0.36	594	O
4	D	74	Olig2 Antibody	NBP1-28667SS	Rb, Poly	motor neuron progenitor cells	500	700	1.4		
5	D	C.13	MAP 2		Rb	pan-neuronal markers	500	180	0.36	488	B
6	D	from Mazzoni lab	HB9		m		100	180	1.8	594	O
7		121	VACHT	MA5-27662	M, Mono	motor neuron marker	500	180	0.36		
8	E	c-28	Purified anti-Tubulin β 3 (TUBB3) Antibody [TUJ1]	801201	m, Mono	pan-neuronal markers					
9			Nkx		m						

2026年2月24日 星期二

- 1. wash with 5 min PBS *3
- 2. 2nd antibody in 10% FBS for 1h RT. 14:30 - 15:30
- 3. wash with 5 min PBS *3
- 4. check staining
- 5. 1ug/mL DAPI staining for 10min
- 6. PBS wash *3
- 7. Store in the cold room

2026年2月27日 星期五

I tried to image IF today, but things didn't work well. I decided to do it next time.

I mount the wells I did IF. However, when I tried to image them with 37C incubator on on new scope, the PFS didn't work (I guess it might because of the extra cover sliip on it). Also, it seems like the cells got spread out along the z axis. It's a bit hard to image cells in the same focal plane. Will try again without incubator next time.

2026年3月6日 星期五

Imaging IF

Old spinning disk confocal

PFS doesn't work, so I took images one by one. Some of the z-stack don't fully work.

SetA OC 60x,704x704, 16bit

	A	B	C	D
1		laser power (%)	exposure time (ms)	
2	DIC			BF
3	405	10	50	DAPI
4	488	50	100	TDP43
5	561	50	500	Importin-b1
6	640	30	100	Halodye (TDP43-Halo)
7	z-stack	0.5 um/step	10	21 steps in total

SetB OC 60x,704x704, 16bit

	A	B	C	D
1		laser power (%)	exposure time (ms)	
2	DIC			BF
3	405	10	50	DAPI
4	488	10	200	RANGAP1
5	561	25	300	Ran
6	640	30	300	Halodye (TDP43-Halo)
7	z-stack	0.5 um/step	10	21 steps in total

fileID_treatment_IF							
	A	B	C	D	E	F	G
1	image	treatment	treat_time	488	561	640	NOTE
2	A6_488-TDP43_561-Impb1_640-Halo_405-DAPI	2.5mM gluc	1h	TDP43	Impb1	TDP43Halo	a half of the z stack of XY012 and XY018 does work.
3	B6_488-TDP43_561-Impb1_640-Halo_405-DAPI	0.15mM gluc	1h	TDP43	Impb1	TDP43Halo	
4	A5_488-RANGAP1_561-Ran_640-Halo_405-DAPI	2.5mM gluc	1h	RanGAP1	Ran	TDP43Halo	
5	B5_488-RANGAP1_561-Ran_640-Halo_405-DAPI	0.15mM gluc	1h	RanGAP1	Ran	TDP43Halo	

Note for MN identity:

C6

The HB9 still stain non-MNs in C6. (XY008 and XY009)

C5

Very few cells have HB9 signal.

non-MNs almost have no MAP2 signal.